



A Multimodal Analysis of a Digital Learning Resource – Number Sense Presented Through Different Modes

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Abstract

Digital learning resources are increasingly used in mathematics education. These learning resources consist of information built on a variety of different modes (i.e., means for communication, e.g., images, written words, mathematical symbols, animations, gestures), which offer the potential for new ways of learning mathematics. This learning resource analysis focused on various modes to present aspects of number sense and what progression exists across a series of subsequent tasks in two exercises. Theoretically, Halliday's metafunctions, Kress's multimodal perspective on communication, and McIntosh et al.'s framework on number sense are used. The findings show that touch is to be seen as an important mode, that different modes function as either intertwined or complementary, and that existing models of progression in number sense have limitations concerning the role of modes. An overall conclusion is that by recognising all types of modes, we could broaden the understanding of how number sense can be communicated. This can provide support for designing learning resources and ultimately support students' learning.

Keywords App · Digital tool · Early years · Learning resource analysis · Mathematics education

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Introduction

There is a rapidly increasing use of digital learning resources in mathematics education internationally, and the digitalization of school practice is a prioritized area in Europe,¹ including Sweden. Digital learning resources include various materials, from single mathematics tasks or activities to more complete programmes that can replace printed mathematics textbooks (Choppin et al., 2014; Swedish Institute for Educational Research, 2017).

The use of digital resources provides potential for embedding multimodal approaches and new ways for students to learn mathematics (e.g., Hoyles, 2018; O'Halloran et al., 2018; Pepin et al., 2017; Usiskin, 2018). Multimodality refers to the interplay between different modes that build the information communicated in the learning resource. Modes are understood as means for communication, such as images, written words, mathematical symbols, and animations (Kress, 2010). The outcome of the use of new means for communicating mathematics depends on the implementation of digital resources in mathematics education (Hoyles, 2018). This, in turn, requires an assessment of the resources' potential for learning mathematics, which is something that is considered difficult by teachers (Pepin et al., 2017).

The area of printed learning resources is well-researched, and analyses of printed textbooks are therefore common, but research focusing on digital learning resources is a newer and more unexplored area, a research area that has grown in recent years to become an important focus in textbook research (Fan et al., 2013; Pepin et al., 2017). The research that does exist on digital learning resources often focuses on developing number sense and often involves conceptual understanding and/or procedural use (Swedish Institute for Educational Research, 2017). Our search found no studies that were related to means for communication and focused on the use of different modes and combinations of these. This article will, through a learning resource analysis, provide new knowledge about digital learning resources from a multimodal approach by analysing a specific app concerning various modes and aspects of number sense. The selected app has been tested and has produced good learning outcomes for students (Judd & Klingberg, 2021), so it is reasonable that the progression built into the app is functional. More knowledge about the app's progression in relation to aspects of number sense and modes will provide support for how learning tools about number sense can be designed or how teaching about number sense can be structured. This will, in turn, give important information about how digital learning resources have the potential to support students' learning.

Background

Three different areas of previous research are relevant to account for the article's focus. Firstly, we describe the role modes have for learning mathematics in general. Then, we describe the role of modes for developing central aspects of number

¹ <https://education.ec.europa.eu/focus-topics/digital-education/actions>

sense specifically. Lastly, we present research focusing on analysing digital learning resources, particularly content analysis.

Modes and Learning Mathematics

The use of different modes is central in mathematics, which is highlighted from different theoretical perspectives, such as cognitive (Duval, 2006), linguistic (Schleppegrell, 2007), or discursive (O'Halloran, 2005). Different modes are, for example, mathematical symbols, writing, images, diagrams, and physical (concrete) objects. Modes are used to create representations, for example, the number five can be represented with five fingers or the symbol 5. *Modes* thus refer to the means used and *representation* refer to what is depicted. Different modes allow students to encounter different perspectives on the same mathematical idea, and a particular idea can be easier to understand in one mode than another (Usiskin, 2018). In particular, combinations of different modes generally contribute to a deeper understanding of a mathematical idea than each mode separately (O'Halloran, 2005). With the introduction of digital learning resources, static images are, for example, replaced or extended with dynamic or interactive images. Written words can be supplemented with, for example, interactive links to videos, symbolic definitions, data for real problems, or tools to perform various mathematical activities. This digital component integration opens new ways to communicate mathematics and construct mathematical tasks.

To learn, understand, and use mathematics, one must be able to interpret information from different modes. Individuals interpret these modes in different ways, forming an overall understanding of the subject (O'Halloran, 2005). Empirical studies show that learning resources in mathematics are challenging for students to read and that this is related to the combination of different modes (Dyrvold, 2016; Norberg, 2023). For example, modes can be contradictory, so information with mathematical symbols can be contradicted by an image or written words (Leijon & Lindstrand, 2012). Particularly, the use of various images can be very demanding for students to handle (Dyrvold, 2016; Jellis, 2008); for example, drawing attention to less relevant parts of an image, leads students to focus on those parts of the exercise rather than the relevant part of the image, focusing on the mathematical content being taught (Sutherland et al., 2001). Research has also shown that the mathematical symbolic language, especially concerning its relation to natural language, creates complexity in the teaching of mathematics (Johansson & Österholm, 2019), and that mathematical texts that include both ordinary words and mathematical symbols require another type of reading than mathematical texts without these kinds of symbols (Jellis, 2008; Österholm, 2006, 2008). These results show that the combination of different modes contributes to the complexity of learning resources in mathematics, and it is argued that students need teaching on how to interact with the learning resource, specifically, how to handle different modes and combinations of them (Norberg, 2023; Segerby & Chronaki, 2018).

In previous research on digital apps, there are studies on children's and students' use of apps that focus on touch and gestures as a means for communication (see e.g., Crescenzi et al., 2014; Janßen et al., 2020; Manshadi, 2022; Sinclair, 2024). Sinclair (2024) focuses on embodiment and distinguishes between actions on the screen and

in the air, arguing for the need for new methods for analysing embodiment. By analysing a child's interaction with a multitouch application focusing on early number sense, the child's use of touch with both fingers, hand, and arm becomes visible. Sinclair describes how the child's use of touch leads to remembering what the child has done previously, ultimately contributing to knowledge. Crescenzi et al. (2014) recognise different types of touch in a preschool setting, namely press, circular, stroke, and tap. They conclude that children's use of digital learning resources, contributes to a broad use of different types of touch. In a case study of two 11-to 12-year-old old students' mathematical meaning making, Manshadi (2022) argues that touchscreen technology has the potential to enhance the connection between mathematical content, modes, and concrete experiences.

In the Swedish Institute for Educational Research (2017) literature review, the potential of digital resources for students' mathematical learning is shown in effect studies, particularly if mathematical concepts and processes are possible to experience both visually and dynamically. In line with the empirical results accounted for above, thus, it cannot be the mere existence of different modes that is problematic, since effect studies show potential for students' learning of mathematics when using digital resources where even more and/or different types of modes are usually used. It is therefore important to investigate *how* different modes present information and what this means for the presentation of the mathematical content. With more knowledge about this, which we produce in this study, we can better understand what learning situations are offered to students when working with digital learning resources.

Modes in Relation to Developing Number Sense

Number sense refers to an understanding of numbers and operations and the ability to use and operate with numbers flexibly in various situations (McIntosh et al., 1992). This includes basic competencies such as being able to communicate the concept of quantity and numbers in various ways, for example, with number words and written Arabic symbols (Elofsson, 2017). Other modes, beyond written words or symbols, are also important concerning number sense, as discussed in the following.

Research implies that people have two inherent core cognitive systems that are involved in representing numbers, which allow us to nonverbally represent numbers without counting and detect differences in magnitude between sets of objects; and represent numbers precisely and quickly identify and keep track of a small number (i.e., 1–4) of individual objects (Dehaene, 2011; Feigenson et al., 2004). Spatial representation of numbers and the development of a mental number line are essential for further successful mathematical development, particularly number processing and arithmetic ability (Booth & Siegler, 2006; Dehaene et al., 2003; von Aster & Shalev, 2007). According to a cognitive study by von Aster and Shalev (2007), the development of number representations follows a four-step hierarchical trajectory: (1) non-symbolic quantity representations, (2) verbal representations, (3) written Arabic symbolic representations, and (4) symbolic mental number line. Moeller et al. (2015) discuss that the development of a mental number line could be encouraged by digital learning resources, particularly if training includes interactions with bodily experiences of numerical concepts, that is, from using numerical symbols to using concrete

objects and non-numerical representations. Furthermore, it is agreed that using the (physical) number line as a didactical tool is powerful when facilitating students' number sense, particularly arithmetic competence (e.g., Blöte et al., 2001; Kilhamn, 2011). Research has, for example, shown that playing games that include a visual-linear representation of numbers improves young students' performance in placing numbers in the right position on a number line, their ability to name numerals, and coming up with the right results for verbal addition tasks (Elofsson, 2017).

Some studies have focused on aspects of progression in students' learning of number sense, where progression refers to the successively sophisticated ways of thinking about a topic (Chen et al., 2017; Lee, 2016). Lee (2016) describes a progression from easy to difficult across various dimensions, which was validated using IRT methodology. Some dimensions are specific to different aspects of number sense, whilst the number of digits serves as a dimension across all aspects of number sense. Different modes are addressed within one specific aspect of number sense: transcoding. This describes a progression from oral verbal language to symbols and then to written verbal language. It also describes the progression regarding the direction of translation/transcoding, where it first occurs *to* symbols and then *from* symbols. Chen et al. (2017) have also validated a model of progression in number sense, based on analyses of previous research and mathematics textbooks. The model describes various pathways of progression, but broadly speaking, the first step involves moving from an understanding of integers to an understanding of fractions and decimals, followed by learning to handle multiple representations, and finally, performing operations with numbers.

From our literature search, it can be concluded that research on the development of number sense in relation to different modes is limited, and few studies focus on modes at all. When they do, they either focus on individual modes and aspects of number sense, such as bodily experience, physical number line, or hierarchy between different modes. This article contributes to more knowledge about the relationship between number sense and different modes. This will broaden the knowledge from a theoretical point of view.

Digital Learning Resource Analyses

Content analyses are relatively common in the research area of mathematics textbooks (Fan et al., 2013), and much of the research has focused on the presentation of specific mathematical topics (e.g., Fan et al., 2018; Johansson & Österholm, 2023; Norberg, 2019; Park, 2016; Wang et al., 2017) or the treatment of mathematical processes (e.g., Bergwall & Hemmi, 2017; Jäder et al., 2020; Lithner, 2004). Gueudet et al. (2018) discuss two ways to analyse mathematical learning resources: on the macro-level, which focuses on the general characteristics of the resource and its role in the educational system; or on the micro-level, which often focuses on opportunities for learning a specific mathematical topic, reasoning, and proof, or mathematical problem-solving. At the micro-level, it is mostly the tasks and activities that form the basis for analysis. The rapid introduction of digital learning resources has expanded research to include, for example, analyses of digital resources' potential for mathematics learning and the appropriateness of various mathematical apps and programs.

Many of these studies include broader content analysis (i.e., on a macro-level), compared to analogue learning resources. For instance, Namukasa et al. (2016) developed an evaluation tool for teachers to analyse apps. The analysis concerned how the curriculum is addressed, the extent to which the app is designed to offer interaction, what type of interactivity is offered to the student, and the graphic quality. Except for the mathematical content, feedback, scaffolding, adaptability, and interaction possibilities can be included in the analysis (e.g., Callaghan & Reich, 2018; Cayton-Hodges et al., 2015; Harrison & Lee, 2018), which is relevant as it affects learning and is not possible to study in the context of analogue textbooks. There are some studies that also include a micro-analysis of the mathematical topic and how this is presented in various ways in the digital learning resource (e.g., Gueudet et al., 2018).

Two studies that address mathematical digital learning resources, based on different modes, are Elia et al. (2023) and Baccaglini-Frank et al. (2020). In a systematic literature review, Elia et al. (2023) establish two different research directions linked to digital technology in preschool. The first point highlights the need for more digital learning resources in mathematics teaching, and the second emphasizes the requirement for clear teacher guides to support these digital learning resources. In a study conducted by Baccaglini-Frank et al. (2020), preschool children's use of two different number sense exercises in an open digital application with multi-touch affordances is analysed. The results revealed that the children used different strategies and a breadth of abilities related to number comprehension, including cardinality and ordinality. Also, embodied interactions enabled the children's engagement and understanding of numbers.

Because digital learning resources are open for new modes compared to analogue textbooks, the introduction of digital learning resources requires new analytical considerations and dimensions (Gueudet et al., 2018). Broadening the approaches to understanding mathematics as a multimodal discourse (O'Halloran et al., 2018) can provide new perspectives on how mathematical content can be presented, which is what this article will contribute to.

Aim and Research Questions

This article aims to contribute to the understanding of how digital learning resources, through their multimodal properties, can support students' mathematical learning, with a delimitation to number sense, using a learning resource analysis. The study addresses properties and progression in two different exercises in a specific app concerning how various modes present aspects of number sense across a series of subsequent tasks in the exercises. The research questions are:

- 1) Which modes exist, and how do the modes present central aspects of number sense?
- 2) What types of progression exist in the exercises, concerning number sense and modes?

Theoretical Assumptions

The theoretical starting points for this study are based on a combination of Halliday's (1978, 2004) metafunctions, Kress's multimodal perspective on communication (2010), and McIntosh et al.'s (1992) framework on number sense. To conduct the study, we proceeded from these three and built our framework based on them, in the following manner: Halliday was used to capture the different functions of the digital resource, while Kress enabled us to capture all means for communication (i.e. different modes). McIntosh et al. made it possible to understand the mathematical content of numbers and operations. These will be described more in the following.

Halliday's metafunctions (1978, 2004) derive from a linguistic perspective where the primary focus is writing and speech. Metafunctions are understood as the different meanings of language in communication. Each metafunction focuses on a specific function of language, *the textual, ideational, and interpersonal metafunction*. There is no hierarchy or order between different metafunctions, but it is about "how the three metafunctions are woven together into the same linguistic unit" (Hasan, 2009, p. 9). The study is based on all three metafunctions as a theoretical starting point, with an emphasis on the textual and ideational metafunctions, given that the study analyses a digital learning resource rather than its use.

In the *textual metafunction* (Halliday, 1978, 2004), the focus lies on the composition of the text, where interest is directed towards what means for communication (i.e., modes) exist, as well as how context is created through the positioning of different information in a text. In relation to this study, this metafunction is about the use of different modes, such as images, symbols, and moving images, which means that Kress (2010) came to use. Since we focus on all means for communication, this made us expand the concept of "text", thus introducing the concept of modes (Kress, 2010). This means that the study is based on a broadened perspective on text, where a text can be composed of several different means for communication, such as written words, images, and mathematical symbols. Together, these form the text. Different modes have different possibilities and limitations, and the different modes together build the information that the learner meets when interacting with the learning resource (Kress, 2010). For example, information communicated through images is not exactly the same information communicated through mathematical symbols.

In the *ideational metafunction* (Halliday, 1978, 2004), language is used to describe the 'reality' of the world, in short, describing what happens. In this article, this meant the mathematical content that was in focus and not the content in a wider sense. Therefore, McIntosh et al. (1992) was used for this part of the analysis.

McIntosh et al.'s (1992) framework for examining basic number sense consists of three main categories: 1. *Knowledge of and facility with numbers*, 2. *Knowledge of and facility with operations*, and 3. *Applying knowledge of and facility with numbers and operations to computational settings*. Each main category is then divided into different subcategories at two levels. This framework is comprehensive and deals with many aspects of number sense that do not constitute content in the analysed app in this study. Therefore, a first explorative analysis of the exercises was conducted to capture which of the subcategories were present. This resulted in a delimitation

of categories, with subcategories of number sense that were relevant to use for the analysis (Table 1).

McIntosh et al.'s framework does not include progression in learning, but only which number sense components are present. It can be assumed that a more in-depth number sense knowledge implies that students master more components within the framework. However, McIntosh has also developed teaching modules based on levels, building on progression of different aspects of number sense, e.g., progression in number range (McIntosh, 2005). It is therefore of interest to study how the learning process can be structured to support progression in students' number sense, also within individual parts or components of number sense.

The *interpersonal metafunction* (Halliday, 1978, 2004) focuses on the relationship between the author and the reader. Since this study does not involve directly studying the user, the focus is on how a student is expected to act when working with the exercise, and how the app invites the reader to engage with the digital resource. Construing the identities and relationships of the participants in the communication is central and whether the author claims, asks questions, offers, or prompts something to the reader.

The theoretical starting points of Halliday's metafunctions (1978, 2004), Kress' multimodality (2010), and McIntosh et al.'s (1992) number sense complement each other in such a way that Halliday makes visible the entire function of the text; Kress provides a broader perspective of the text, including all recourses used for communication (modes), and McIntosh et al. provide tools for capturing the mathematical content, which is connected to the ideational metafunction. This combination of per-

Table 1 Relevant categories with subcategories, from McIntosh et al. (1992)

1. Knowledge of and facility with numbers
1.1 Sense of orderliness of numbers
1.1.1 Place value (e.g., patterns 1,2,3 ..., and 21, 22, 23 ..., and 31, 32, 33)
1.1.2 Relationship between number types (e.g., $\frac{3}{4}$ is a fraction less than 1)
1.1.3 Ordering numbers within and among number types (e.g., 4 is a number between 2 and 5)
1.2 Multiple representations for numbers
1.2.1 Graphical/symbolic (e.g., $4+4+4$ is the same as 3×4)
1.2.2 Equivalent numerical forms, incl. decomposition/recomposition (e.g., $25+28=25+25+3$)
1.2.3 Comparison to benchmarks (e.g., $\frac{5}{8}$ is "a bit more than $\frac{1}{2}$ ")
2. Knowledge of and facility with operations
2.1 Understanding the effect of operations
2.1.1 Operating on whole numbers (e.g., $2+1=?$)
2.1.2 Operating on fractions/decimals (e.g., $\frac{1}{4}+\frac{1}{4}$)
2.2 Understanding mathematical properties
2.2.1 Commutativity (e.g., $2+7=7+2$)
2.2.2 Associativity (e.g., $(8+6)+4=8+(6+4)$)
2.2.3 Distributivity (e.g., $4 \times 36=(4 \times 35)+(4 \times 1)$)
2.2.4 Identities (e.g., $2+1=3$; $3=2+1$)
2.2.5 Inverses (e.g., $4+3=7=>7-3=4$)

spectives broadens the possibility of understanding how the app makes use of different modes in relation to number sense.

Method

The study is based on a learning resource analysis, in which the research team conducted a joint, structured analysis of the digital app. The digital learning resource Vektor has been shown to contribute to increased knowledge among students who regularly use the resource (Judd & Klingberg, 2021). Based on this, we chose to analyse this particular app to gain an understanding of how it is constructed with a delimitation to number sense and modes. Vektor is developed by the Cognition Matters Foundation² and is downloaded to a tablet. Vektor is structured like a game with different exercises focusing on different mathematical content. Each exercise consists of a number of rounds, and each round consist of a number of tasks. To level up in the game, the student must complete rounds of tasks in the exercises without errors. The selection of exercises for the analysis was based on the inclusion of number sense. This criterion was fulfilled by two of the exercises, called Number line (Fig. 1) and Number pals (corresponding to what is commonly referred to as number bonds, Fig. 2).

In the Number line exercise, the student receives a number or a calculation and produces answers by dragging the finger on the number line to represent the number or calculation. When an answer is completed, a new number or calculation appears on the screen. A round consists of seven such tasks, which are marked with dots in the upper left corner. A beep will sound when the answer is correct, and a wrong answer is documented by filling out one of the three dots in the upper right corner. After 3 errors, the student receives help from a hand that shows how to drag on the number-line. In the Number pals exercise, the student builds number bonds of red and green rectangles with corresponding numbers. The student selects the bar with the numbers of green rectangles to fill up the bar with the red rectangles by dragging this “green bar” to the “red bar” and thereby fill the whole bar. When one bar is completed, a new bar is shown above the first one until the entire left field is filled. Thus, each round consists of eleven tasks. Each round is on time, and a wrong answer is shown with a “shake” of the picked bar and then a bounce back to its position in the right box.

In the analysis, we started from a bottom-up perspective on how the exercises change and relate this to metafunctions (Halliday, 1978, 2004), modes (Kress, 2010) and aspects of number sense (McIntosh et al., 1992). We started by exploring the two exercises concerning the properties of the app and the exercises, providing both correct and incorrect answers to the tasks, in order to be familiar with the exercises. This was documented. After that, the analysis of the exercises was divided into two parts answering the two RQs:

- 1) *Analysis of which modes exist and how the modes present central aspects of number sense (to answer RQ1).* First, we documented which modes were used in the

²The foundation no longer exists.

Fig. 1 Example of the exercise Number line (All images are shown with permission from the Cognition Matters Foundation.)

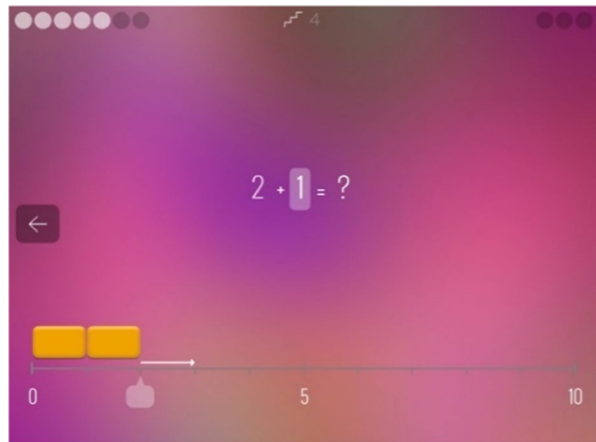
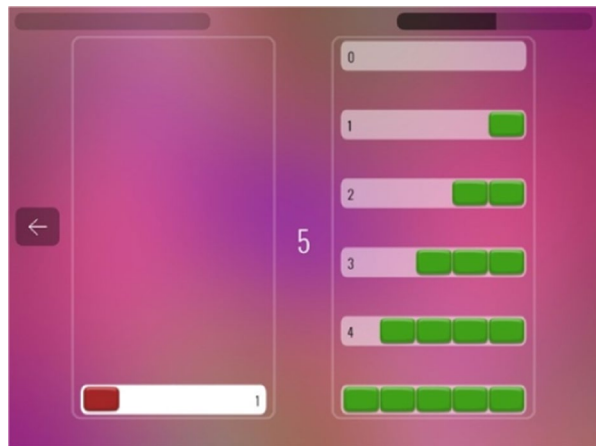


Fig. 2 Example of the exercise Number pals



different exercises. For this Halliday (1978, 2004) was used to capture how the modes should be used (i.e., what functions they were given) in the exercises, Kress (2010) was used to capture which modes existed, and McIntosh et al. (1992) was used to capture which aspects of number sense were in focus. Departing from Halliday's three metafunctions described above (theoretical assumptions), the following questions were asked for each round in the respective exercise:

- Which modes exist, and what do the different modes present? (Textual function)
- Which aspects of number sense does the exercise cover? (Ideational function)
- How is a student expected to act when working with the exercise? (Interpersonal function)

In a spreadsheet, for each round in each exercise, the following was documented: the first-level subcategory of number sense, the second-level subcategories, which

modes presented the respective subcategories, and how this was done (see Table 2). A thorough example of this step is given below.

in the different rounds

Analysis of what types of progression exist concerning modes and number sense (to answer RQ2). In this part of the analysis, we searched for patterns in the different rounds of the exercises, where each round was compared to the previous one and changes between rounds was documented. Since the analysis is based on capturing the progression of the app it is based on answering all tasks in the round correctly. Previous research shows that progression is linked to types of numbers, performing operations with numbers and the use of different modes (Chen et al., 2017; Lee, 2016). Thus, it is relevant to analyse differences between each round to see how the app produces more sophisticated ways of thinking (cf. Chen et al., 2017; Lee, 2016). This was documented in a spreadsheet to discern patterns regarding numbers, operations, and modes in succeeding rounds in each exercise. This provided information on how each exercise progressed with respect to mathematical content and modes. The progression for each exercise was compared to the other to capture an overall progression.

Next is an example of step one in the analysis procedure. The analysis of the first round in Number pals (Fig. 3) gave the following results.

- *Textual function* (the existing modes and what information they present) (Halliday, 1978, 2004; Kress, 2010): Existing modes are images, mathematical symbols and touch. This round of the exercise consists of images in the form of different numbers of red (start from the left) and green (start from the right) rectangles placed in white and grey "bars" of the same size. Each bar is full when it consists of five rectangles. These bars are in turn, placed above each other in boxes. There are six bars in the right box, with the bottom bar filled with five green rectangles, the next one with four green rectangles, and so on, and the top bar does not include any green rectangles. There is also a mathematical symbol in the form of the numeral "5" placed in the middle of the screen, between the two boxes. The colour white of the bar with red rectangles shows "activity", and the colour changes to grey when it has been combined with an exact number of green rectangles so that the bar is filled. Touch is used when dragging the bars.

- *Ideational function* (the mathematical content) (Halliday, 1978, 2004): The exercise is about combining different numbers of coloured rectangles to fill a bar of a specific size. The size of the bar corresponds to the number symbolised by the numeral in the middle of the screen. This is interpreted as focusing on the second-level subcategories *decomposition/recomposition* (1.2.2, Table 1) and *commutativity* (2.2.1, Table 1) (McIntosh et al., 1992).

- *Interpersonal function* (the relationship between the exercise and the user) (Halliday, 1978, 2004): The student is offered a bar with a number of red rectangles in the left box, the symbol 5 in the middle, and bars with different numbers of green rectangles in the right box. The task is to select the bar with the number of green rectangles that will fill up the bar with the red rectangles and drag this "green bar" to the "red bar" to combine these two. If the wrong bar with green rectangles is selected, the red rectangles "shake" and the bar with the green rectangles bounces back to its

position in the right box. The student can make as many attempts as she likes until the time for the exercise has run out.

Results

The result is divided into two sections, answering each research question: describing which modes exist, and how the different modes present central aspects of number sense (RQ1), and presenting what types of progression exist in the exercises, concerning number sense and modes (RQ2).

Existing Modes and Central Aspects of Number Sense

Under this heading, results are first presented that identify which modes exist in the exercises, how different modes present central aspects of number sense and then results on how the combination of different modes relates to aspects of number sense.

How Different Modes Present Central Aspects of Number Sense

The results show that the existing modes in the two exercises are images, mathematical symbols and touch. Touch is a mode that becomes visible when interacting with the exercise and can be described as the movement that is performed by the learner, for example, when dragging an image from one place to another. Table 2 shows the types of modes and central aspects of number sense in the exercises. In this section, a presentation of the results on how the modes present central aspects of number sense is given. As seen in Table 2 below, all three identified modes, images, touch, and mathematical symbols, appear in relation to all aspects of number sense in different variations. There are eight different types of images (I1-I8), four different types of touch (T1-T4), and three different types of mathematical symbols (M1-M3). In total, each mode addresses all four aspects of number sense but to varying degrees with an emphasis on Second aspect: 1.2 Multiple representations for numbers. See Table 2 for more detailed information on this but also to gain insight into the analysis work and thus aspects of transparency. The results also show that the Number pals exercise focuses on cardinality while the Number line exercise focuses on ordinality.

How the Combination of Different Modes Relates to Central Aspects of Number Sense

The results show that the interaction of different modes has either an *intertwined* or a *complementary* function in relation to number sense. If the modes are *intertwined*, this means that one mode cannot be used without the other, since they only exist together. An example of this is shown regarding orderliness (first aspect 1.1) and touch- and image-mode in exercise Number line (Fig. 4). Touch is shown in the movement when solving the exercise, which is done by dragging with the finger to create the number that is demanded. The image appears in the movement, as an arrow. So, one mode does not emerge without the other. In contrast to the intertwined modes, the results

Table 2 Aspects of number sense presented in images, touch, and mathematical symbols (building on Halliday, 1978, 2004; McIntosh et al., 1992)

	First aspect: 1.1 Sense of orderliness of numbers - Concerns ordering numbers within and between number types	Second aspect: 1.2 Multiple representations for numbers - Concerns numbers in different contexts and how they can be expressed in different ways	Third aspect: 2.1 Understanding the effect of operations - Concerns operating with numbers in addition and subtraction	Fourth aspect: 2.2 Understanding mathematical properties - Concerns commutativity and associativity
Images				
11	Combinations of boxes in different colours (Number pals) Example: boxes that visually can be combined with other boxes to fill out the row, like pieces in Tetris that fit together and fill the row	x		
12	Dividing line (5 and 5 or 10 and 10) (Number pals) Example: the number ten is presented with a dividing line in the middle of the row dividing 10 into 5 and 5	x		
13	Part of a whole (Number line) Example: an image with three of four squares coloured showing the number $\frac{3}{4}$	x		
14	Intervals on a number line (Number line) Example: an interval in the shape of an arrow on the number line	x		
15	Number line (Number line) Example: a number line with tick marks for each number where every fifth mark is slightly longer	x		
16	Numbers of boxes ordered from 0–5, 0–10 etc. (Number pals) Example: concerning the number five, the boxes to the right are ordered: 0, 1, 2, 3, 4 and 5 from the top of the bottom in the right column	x		
17	Combinations of different numbers of boxes and an interval on the number line (Number line) Example: two boxes lined up on a number line, from 0 to 2 on the number line	x		
18	Combinations of different numbers of boxes in different colours (Number pals) Example: three red boxes are added to two green boxes (here is the number of boxes in focus)	x	x	x
Touch				

Table 2 (continued)

	First aspect: 1.1 Sense of orderliness of numbers - Concerns ordering numbers within and between number types	Second aspect: 1.2 Multiple representations for numbers - Concerns numbers in different contexts and how they can be expressed in different ways	Third aspect: 2.1 Understanding the effect of operations - Concerns operating with numbers in addition and subtraction	Fourth aspect: 2.2 Understanding mathematical properties - Concerns commutativity and associativity
T1	Boxes being combined when performing the calculation. (Number pals) Example: boxes that are combined with other boxes by dragging boxes from the right column to boxes in the left column to fill out the row (like pieces in Tetris that fit together and fill the row)	x		
T2	Combinations of different number of boxes in different colours when performing the calculation (Number pals) Example: two green boxes are dragged from the right column to three red boxes in the left column (here is the number of boxes in focus)	x	x	x
T3	Combinations of different numbers of boxes and an interval on the number line shown when performing the calculation (Number line) Example: when performing the calculation $2+4$ two yellow boxes are lined up on a number line when dragging with the finger on the number line from 0 to 2 and four more when dragging from 2 to 6. It is also possible to do the calculation $4+2$	x	x	x
T4	Intervals on a number line shown when performing the calculation. (Number line) Example: when performing the calculation $1/4 + 2/4$ an arrow from 0 to $3/4$ is shown when dragging with the finger on the number line from 0 to $3/4$. It is also possible to do the calculation $2/4 + 1/4$	x	x	x
Mathematical symbols				
M1	A position on a number line (Number line) Example: numbers on the number line are marked with a symbol	x		
M2	A number type: Natural numbers (Number line+Number pals) and Rational numbers (Number line) Example: "5", "12", "3/8"	x		
M3	Addition- or subtraction operation with two or three terms (Number line) Example: " $3+2=$ ", " $2+3=$ ", " $2-2+1=$ "	x	x	x

Fig. 3 Number pals, Learning goal: number 5, round 1

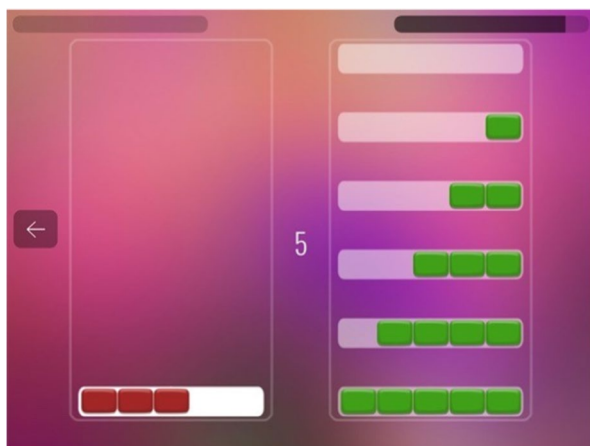
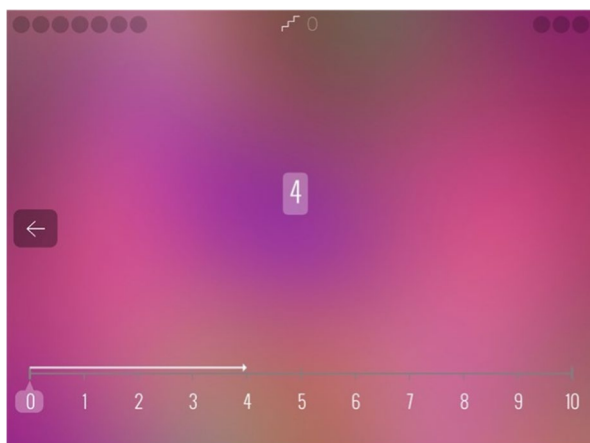


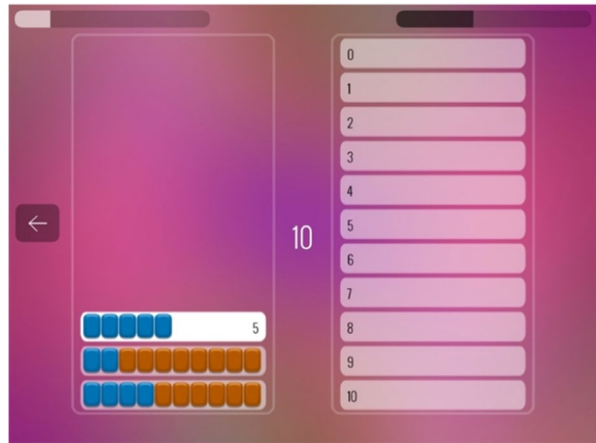
Fig. 4 Number line, Learning goal: natural numbers 0–10 Round 0



show that modes could also be *complementary*. This means that one mode exists regardless of another mode. An example of this is shown in mathematical symbols-mode. Here, symbols present a position on the number line. The number line can be used without the symbols, as the numbers are shown with tick marks for each number and the mathematical symbol thereby complements the image. The mathematical symbol gives information usable in the navigation on the number line. As a general result description, it should be noted that when the exercise presents operations, the results are not presented in mathematical symbols, only in images and touch.

Another example of an intertwined or complementary function is given regarding operations (third aspect 2.1) and exercise Number pals (Fig. 5). Again, touch and image are intertwined in that the right-hand column initially shows only number symbols in the rows. When you then drag the row to the left column and drop it, yellow boxes appear, filling the row with blue and yellow boxes (like the two bottom rows in Fig. 5). This means that touch is used to perform the calculation. Without the use of touch, two numbers are made visible but not the operation. The complementary

Fig. 5 Number pals, Learning goal: number 10, Round 13



function is shown in the left-hand column, where the number is given in both blue boxes and symbols.

Summary

Images, touch, and mathematical symbols appear in both exercises consistently. The images present the number visually. This is strengthened by different colours in Number pals to visualise which number makes up a whole ($2 + 3 = 5$). The touch appears throughout the exercises and is important for presenting the numbers and operations. It is the touch that makes the operations visible, and the movement is closely related to the calculation. The mathematical symbols are, of course, essential when presenting numbers, but other modes also present aspects of number sense, as seen above. This is, for instance, seen in Number line where the number “4” is presented not only in the symbol but also in an image in the shape of an arrow on the number line and in the touch when using a finger to drag and perform the number. The different modes have either an intertwined or complementary function. As previously mentioned, if the modes have an intertwined function, one mode depends on another and cannot exist alone, like touch needs an image to exist in these two exercises. If the modes have a complementary function, the modes can stand alone and present information by themselves. This is seen when, for instance, the different numbers to choose from are presented in both image-mode and mathematical symbols-mode in Number pals.

Progression, Regarding Modes and Number Sense

In this part of the results, different aspects of progression are presented, which are progressions linked to numbers, operations, and modes, highlighting two of the metafunctions: the mathematical content (ideational metafunction, Halliday, 1978, 2004), and the existing modes (the textual metafunction, Halliday, 1978, 2004). The number range is the dominant aspect of the progression in both exercises. The exercises move from lower to higher number ranges and from natural to rational number sets.

Operating with numbers is the second aspect of progression, which concerns both the amount of numbers in the operation and the kind of operation. The exercises move from including only single numbers to including operations with, first two numbers and then three numbers, as well as from addition to subtraction. The third aspect of progression concerns the existence of image-mode and mathematical symbols-mode, where the exercises change from including a few mathematical symbols and more images to the opposite.

In this section, we describe how the progression occurs in the two exercises, first regarding Number line and then regarding Number pals. Then a section with identified progression and overall conclusions based on the analysis of both exercises is presented, followed by a short summary of the section. Each description is introduced with a table showing the different aspects of progression found, followed by a description of more detailed results. The description of the detailed results follows the structure of the table for the different columns from left to right and the rows from the top to the bottom. Here again, the tables also serve the function of providing insight into the analysis work and thus aspects of transparency.

Progression in Exercise Number Line

As mentioned earlier, the number ranges dominate the progression. To this, the existence and amount of numbers (one, two, or three), operations of arithmetic (addition and subtraction), and the graduation of the number line (all whole numbers, every fifth, every tenth number) are added. Apart from this, the existence of the image-mode is part of the progression. Table 3 shows the progression in Number line concerning number range (columns 1–3), the existence of mathematical symbols mode (column 4), amount of numbers (columns 5–7), arithmetic operations (columns 8–9), numberline grading (columns 10–14) and existence of image mode (column 15). The presentation of the exercise in Table 3 follows the structure of the number ranges and is divided into Sects. 1–3. It is notable that when the exercise changes and gets more demanding according to one aspect (number range, amount of numbers, operations of arithmetic, or graduation of the number line), the rest of the aspects are kept the same or get more supportive. For instance, when the exercise changes to a higher number range, other aspects of the exercise get more detailed, like a more graduated number line.

In Sect. 1 (rounds 0–22), there are initially single numbers between 0–10, in mathematical symbols, and a graduation of 0–10 on the number line. The symbols on the number line are then removed gradually. When the exercise changes to addition, the image-mode is added. The next change is a switch to subtraction instead of addition (see example in Fig. 6), following the same pattern as in addition. Then the two operations of arithmetic are mixed, and after that, operations with three numbers are presented.

In Sect. 2 (rounds 23–46), the number range is increased to 0–20. The layout of the exercise follows in many ways the layout of Sect., with one exception: the graduation of the number line is more stable and only changes in the end. In Sect. 3 (rounds 47–53), rational numbers are presented. Notably is that no mathematical symbols exist at the beginning of this part of the exercise, but images (Fig. 7). Here, the number line is graded from $0/4$ – $4/4$ (quarters) or $0/8$ – $8/8$ (every eighth).

Table 3 Progression Number line (building on Halliday, 1978, 2004; McIntosh et al., 1992)

Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Aspect															
Round	Number range 0–10 (Natural numbers)	Number range 0–20 (Natural numbers)	Number range 0–1 (Rational numbers)	Mathematical symbols mode:	One number	Two numbers	Three numbers	Addition	Subtraction	Number line grad-ing: 0–10	Number line grad-ing: 0, 5, 10 etc	Number line grad-ing: 0, 10 etc	Number line grad-ing: 0/4–0/8	Number line grad-ing: 0/4–0/8	Image-mode: Boxes, arrows, fractions in images
Sect. 1 0–1															
	2														
	3														
	4														
	5–10														
	11–12														
	13–15														
	16														
	17–19														
	20–22														
Sect. 2 23–25															
	26–32														*
	33–38														*
	39–42														*
	43–46														*
Sect. 3 47–50															
	51–53														*

* No data is available from rounds 23–46 because the application was shut down

Fig. 6 Number line, Learning goal: natural numbers 0–10 Round 11

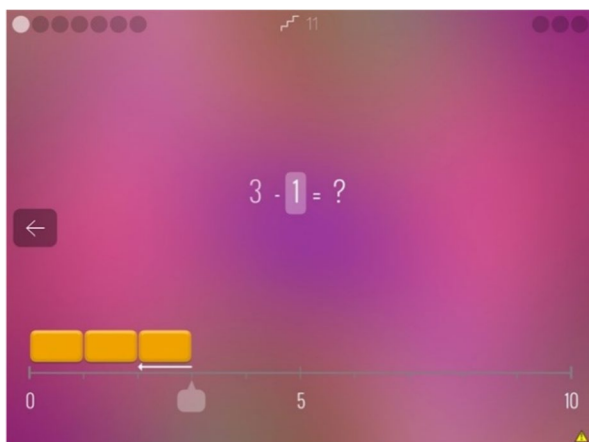
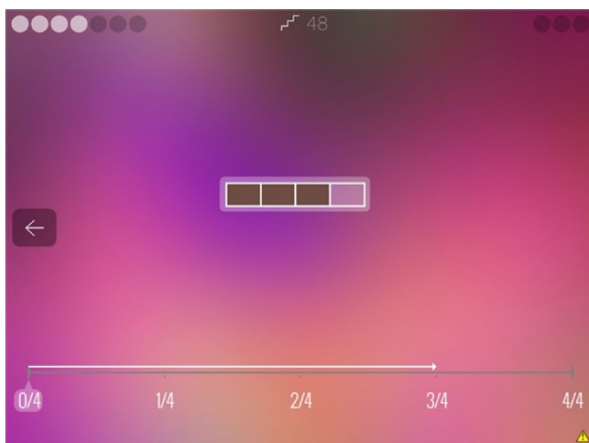


Fig. 7 Number line, Learning goal: rational numbers 0–1, Round 48



Progression in Exercise Number Pals

In this exercise, the number range progression is combined with the existence of mathematical symbols-mode, dividing lines, and the existence of image-mode (boxes in different colours). Table 4 shows the progression in Number pals concerning the number range progression (columns 1–4), the existence of mathematical symbols-mode (column 5), dividing lines (column 6), and the existence of image-mode (columns 7–9). The presentation of the exercise in Table 4 follows the structure of the number ranges and is divided into Sects. 1–5. The exercise changes gradually, presenting a larger number with less existence of information in image-mode such as images and dividing lines.

In Sect. 1 (rounds 1–8), the number 5 is presented, first done with boxes on both the right and left sides. Then, mathematical symbols are added, and gradually the boxes are taken away. First, the boxes are shown on the right side (from where the user should select), and then there are no boxes, but the boxes will be shown when the correct pair of numbers is chosen. In Sect. 2 (rounds 9–20), the number 10 is focused

Table 4 Progression Number pals (building on Halliday, 1978, 2004; McIntosh et al., 1992)

Column	1	2	3	4	5	6	7	8	9
Round	Aspect								
	Num- ber 5	Num- ber 10	Num- ber 15	Num- ber 20	Math- ematical symbols- mode	Image- mode: Divid- ing lines	Image- mode: Boxes on the left side	Image- mode: Boxes on the right side	Image- mode: Boxes vis- ible when right and left is paired
Sect. 1	1								
	2–3								
	4								
	5–8								
Sect. 2	9								
	10–11								
	12								
	13–17								
	18								
	19–20								
Sect. 3	21					2			
	22					2			
	23					1			
	24								
	25								
	26–30								
Sect. 4	31								
Sect. 5	32					3			
	33					3			
	34					1			
	35								
	36								
	37–40								

on (see example in Fig. 8). The progression is partly the same as for the number 5, with two exceptions: there is a dividing line showing 10 as $5 + 5$, appearing in the first rounds, and the section with the number 10 does not end with presenting the numbers only in symbols, like when 5 was focused on. When 10 is presented, there is an extra round where both the dividing line and boxes are shown.

In Sect. 3 (rounds 21–30), the number 15 is focused on (see example in Fig. 9). First, there are two dividing lines ($5 + 5 + 5$) and then one ($10 + 5$). The section ends, like when practicing the number 5, without boxes, but boxes are shown when the right pair of numbers is chosen. In Sect. 4 (round 31) returns to number 10 and is presented with a dividing line and no images, except the images shown after the calculation. In Sect. 5, (rounds 32–40), the number 20 is presented in the same way as the number 15, but first with three lines ($5 + 5 + 5 + 5$) and later with two ($10 + 10$).

Fig. 8 Number pals, Learning goal: number 10, Rounds 10–11

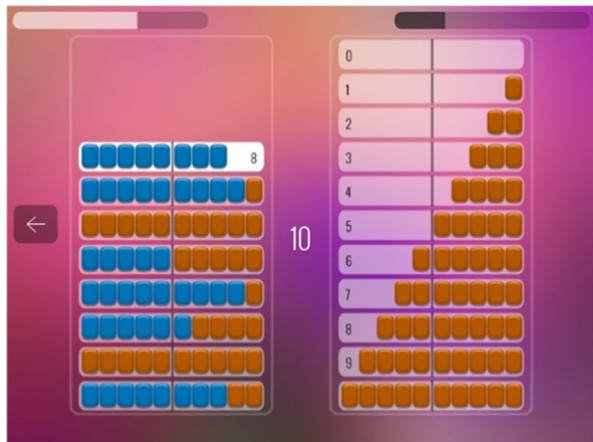
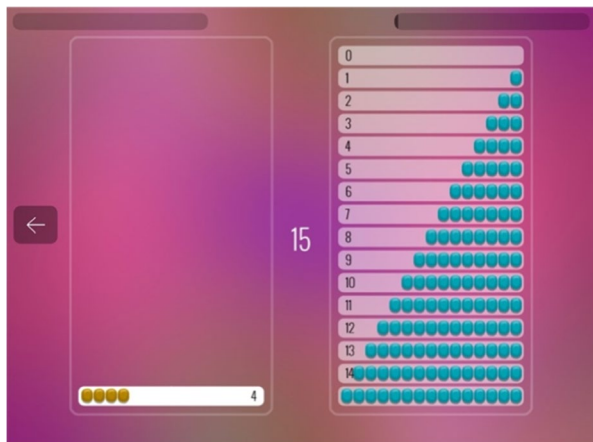


Fig. 9 Number pals, Learning goal: number 15, Round 24



Progression in Both Exercises – Overall Conclusions

Based on the descriptions of progression in the two exercises, we here focus on what the progression looks like in the various aspects related to (1) numbers, (2) operations, and (3) modes.

Numbers: From Lower to Higher Number Ranges; From Natural to Rational Numbers

The first aspect of mathematical progression concerns numbers in two different ways: (1) from lower to higher number ranges, and (2) from natural to rational numbers. Number pals start with numbers up to 5 and Number line with numbers up to 10, which are then increased. In the exercise, Number lines with natural numbers are then followed by exercises with rational numbers.

Operations: From Single Numbers to Operations, From Addition to Subtraction

The second aspect of mathematical progression concerns operations in two different ways: (a) from single numbers to operations with numbers, and (b) from addition to subtraction. Both exercises focus on single numbers but in different ways. Number line focuses on creating the numbers along a number line. In Number pals, two numbers that together correspond to the requested number are paired, and, in this exercise, there is no progression concerning operations. In Number line, the progression moves to operations with numbers in symbols, first with two numbers and then three. When it comes to the operations of arithmetic, addition is first presented, then subtraction, and then addition and subtraction are mixed.

Modes: From Images to Mathematical Symbols

The third aspect of mathematical progression concerns modes. In the first rounds images express the mathematical content. Arrows, boxes, and lines express numbers or operations. Exercises are presented without using mathematical symbols. In Number pals, this is more clearly part of the layout, while this is shown only when presenting rational numbers in Number line. In later rounds there are fewer images, and images are changed into mathematical symbols. The existence of different modes, from images to mathematical symbols, happens gradually. The content is based more on images and variations of images in Number pals than in Number line. The existence of the dividing line is similar and is prominent at the beginning of a new number range, with one, two, or three dividing lines, and is gradually removed.

Summary

In summary, according to different aspects of progression, the layout of the exercises could be understood as using images and smaller natural numbers is more basic mathematics, and using larger numbers, mathematical symbols, and rational numbers in operations with three numbers and subtraction is more advanced. When it comes to the existence of different modes, the images are prominent at the beginning of the exercises and could be described as Multiple representations of numbers (1.2) and Sense of orderliness of numbers (1.1). This means that we can talk about number sense even though no mathematical symbols are shown. For instance, at the beginning of the exercise Number pals, coloured boxes appear when presenting numbers without using symbols. So initially, images have a prominent position, and as the exercise continues, the mathematical symbols are more crucial.

Discussion and Conclusion

The study reported in this article produced results concerning the understanding of how digital learning resources, through their multimodal properties, can support students' mathematical learning, with a delimitation to number sense. Drawing on these results, the discussion focuses on the following aspects: (1) the touch mode as support for experiencing mathematics; (2) mathematical content is communicated differently depending on whether the modes have an intertwined or complementary function; and (3) the role of modes in number sense progression.

The results show that touch is an important aspect of the exercises, supporting mathematical experiencing, and a mode in itself. It is in the movement that the mathematical content is made visible. This is important to elucidate, as the touch mode could otherwise be overlooked, and the exercises would then be understood as consisting only of static modes (images and mathematical symbols). Touch is not just a means of reaching an answer in a calculation, a way to present mathematical content in an act, but a goal itself. Touch makes it possible to communicate something dynamic, for instance, adding or taking away. Through touch, the student is supported in getting an idea of what the operation means when moving the finger along the number line and when pairing boxes. This movement can support the understanding of the calculation (subtraction or addition). Pairing together a number of boxes with another number of boxes contributes to the student's understanding of the current number, and that there are multiple ways of presenting numbers and operations with numbers. This can be compared to moving a finger along a physical number line, which has been proven supportive for students' development of number sense (e.g., Blöte et al., 2001; Kilhamn, 2011). Also, Moeller et al. (2015) highlight that bodily experiences of numerical concepts can support the development of a mental number line, which is ultimately crucial for developing good number sense (Booth & Siegler, 2006; Dehaene et al., 2003; von Aster & Shalev, 2007). Previous research describes touch as a key aspect for understanding the learning of mathematics, through various types of analysis, such as those focusing on different types of touch (Crescenzi et al., 2014) or as part of embodiment (Sinclair, 2024). Sinclair concludes that touch contributes to remembering and, by extension, learning. This could possibly imply that using a finger to draw on a number line could be more beneficial for learning than writing a mathematical symbol on a line or pressing a key. Crescenzi et al. (2014) also highlight the advantages of digital learning resources, as the children then make broad use of different types of touch. Our study has contributed with a different type of analysis concerning touch mode, through a more detailed analysis of the relationships between different modes and specific mathematical content. This analysis reinforces the conclusions from previous research regarding the importance of touch as a mode for understanding the learning in mathematics.

Another important result was that the relationship between the modes in the exercises could be understood as either intertwined or complementary. If the modes are intertwined, one mode cannot be used without the other, and they only exist together. For instance, when performing the calculation in the analysed exercises, this is done

by dragging with the finger to create the number that is demanded. The touch mode is only visible in the movement and is intertwined with the image-mode. Prior research has highlighted that the combination of different modes can contribute to a deeper understanding of mathematical ideas (e.g., O'Halloran, 2005). Our results contribute with a more detailed description of how this combination can have different characters, and how this can relate to the students in different ways. If modes are intertwined, then a combination of different modes is presented that could imply that a more suitable learning situation could be offered to the student, since the different modes are tightly connected to each other. The other option shown in the results was when modes were complementary. Then one mode exists regardless of another mode, for instance, in the number line exercise, the mathematical symbols-mode would present a number positioned on the number line even without the image-mode. It is, of course, possible to combine and focus on different complementary modes at the same time in the exercise, but this is an endeavour that must be more actively chosen by the user, in contrast to intertwined modes. This is important for both teachers and textbook authors to be aware of. The mathematical content is communicated differently depending on whether the modes that present the content have an intertwined or complementary function. This result is important for mathematics teaching because it is made visible that modes communicate mathematics differently, and with this knowledge more conscious learning situations can be offered to students. In addition, it is an interesting empirical question whether and how these different ways could affect students' work with the exercise.

The third aspect to discuss is the role of modes in number sense progression. There are some similarities between our results and prior research on progression in number sense, concerning types of numbers and operations. Our results show a progression from smaller to greater numbers and from natural to rational numbers. This is similar to the progression in number of digits from Chen et al. (2017) and the progression from integers to fractions and decimals from Lee (2016). Also, our results show a progression from single numbers to operations and from addition to subtraction. This is similar to the progression in first understanding numbers and then operations on numbers from Chen et al. (2017). However, concerning the role of modes, there are both some similarities and some differences between our results and prior research on progression in number sense. The similarity concerns a progression towards the use of symbols, when images is a starting point in our results, while non-symbolic quantities and verbal language is a starting point in von Aster and Shalev (2007) and verbal language is a starting point in Lee (2016). Although our results show a progression from images to symbols, our more detailed analyses of how different modes are used in the exercises also show how these are deeply integrated with each other and with the mathematical content. This integration is not captured in the models of progression of number sense presented in previous research. In particular, Lee (2016) has transcoding as a separate component of number sense, which is not related to other aspects of number sense, and Chen et al. (2017) describe the handling of different modes as a separate step in the progression. These models are therefore unable to capture the complexity found in real-life situations when students work with numbers. A challenge for future research is therefore to develop models of progression that are

more integrated in terms of how modes relate to each other and to mathematical content, where the handling of modes should not be described as a separate part of number sense and not be based on linear models involving a separation of different modes.

In conclusion, the study reported in this article has shown the importance of making visible all the different modes that together form the learning resources, and that this should be related to a specific mathematical content. Our results show that touch as a mode has the potential to be supportive when learning number sense, reinforcing similar conclusions from other types of studies in previous research. Moreover, we have shown how modes have either an intertwined or complementary function, which affects how the mathematical content is communicated and, by extension, what students learn. McIntosh et al. (1992) describe number sense “as a propensity for and an ability to use numbers” (p. 4). This was written over 30 years ago, and today we have a wider focus on number sense that includes multimodality. Our study shows that modes and number sense should not be separated but are closely related. The use of different modes to communicate number sense could thereby be included in McIntosh et al.’s framework. With this expanded perspective on number sense, students’ knowledge could be understood in a more nuanced way. For example, we have shown how existing models on progression of number sense need to be developed to take into account the more complex relationships that exist between modes and number sense. By recognising all types of modes in relation to number sense, the conceptualisation of number sense can be broadened, for a more holistic and detailed perspective on students’ learning of number sense.

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Data Availability Due to the fact that the foundation that developed the learning resource analysed no longer exists, the data is not available.

Declarations

Competing interests The authors have no competing interests to declare that are relevant to the content of this article.

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